

The signed Jacobian integral is a sign times target volume

MathlibAnnex.BilipschitzOrientation.integral_det_fderiv_eq_signed_volume

theorem

Uses preconnectedness to obtain a constant, positive measure to identify its sign, and finite measure to integrate it.

Statement

Let $n \in \mathbb{N}_0$ and let $E = \mathbb{R}^n$ have the sup norm $\|\cdot\|_\infty$. Let $S, T \subseteq E$ be open, and let $f, g : E \rightarrow E$ be globally Lipschitz maps whose restrictions $f|_S : S \rightarrow T$ and $g|_T : T \rightarrow S$ are mutually inverse. Fix a number $\ell > 0$ such that

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \quad (x, x' \in E).$$

Assume that T is preconnected (it has no separation into two nonempty relatively open subsets) and that its Lebesgue measure $\lambda(T)$ satisfies $0 < \lambda(T) < \infty$.

Write $Df(x) : E \rightarrow E$ for the Fréchet derivative of f at x when it exists, and set $Df(x) = 0$ at other points. Define the signed Jacobian by $J_f(x) = \det Df(x)$. Then there exists $\varepsilon \in \{-1, 1\}$ such that

$$\int_S J_f(x) d\lambda(x) = \varepsilon \lambda(T).$$

Here the finite measure $\lambda(T)$ is identified with its real value, so this is an equality of real numbers.

Assumptions

The target T is **preconnected**, has **positive** Lebesgue measure, and has **finite** Lebesgue measure:

$$T \text{ is preconnected, } \quad 0 < \lambda(T) < \infty.$$

These conditions are additional to openness and the following inverse and global metric assumptions.

The sets $S, T \subseteq E$ are open. The inverse and image conditions are

$$\begin{aligned} f(S) &\subseteq T, & g(T) &\subseteq S, \\ g(f(x)) &= x \quad (x \in S), & f(g(y)) &= y \quad (y \in T). \end{aligned}$$

The global bounds use specified constants $C_f, C_g \geq 0$ and $\ell > 0$:

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \leq C_f \|x - x'\|_\infty \quad (x, x' \in E),$$

$$\|g(y) - g(y')\|_\infty \leq C_g \|y - y'\|_\infty \quad (y, y' \in E).$$

The estimates hold on all of E , not just on S or T . The inverse identities are required only on S and T .

Conclusion

The integral of the signed Jacobian is either $\lambda(T)$ or $-\lambda(T)$. The sign ε is obtained from the maps; it is not fixed in advance.

The proof uses the three target hypotheses at different stages: preconnectedness gives an almost-everywhere constant, positive measure identifies its value as a sign, and finite measure permits integration of that constant.

Proof route

Identify the almost-everywhere constant value of the transported sign at a point outside its null exceptional set. Then substitute the constant function 1 into signed transfer.

▼ Proof steps

1. **Obtain a constant and identify its value.** Define

$$s_f(x) = \begin{cases} 1, & J_f(x) \geq 0, \\ -1, & J_f(x) < 0, \end{cases} \quad \sigma_f(y) = s_f(g(y)).$$

By constancy of the transported sign, applied with the assumed preconnectedness of T , there is $c \in \mathbb{R}$ for which the set

$$N_c = \{y \in T : \sigma_f(y) \neq c\}$$

satisfies $\lambda(N_c) = 0$. Since $\lambda(T) > 0$, choose $y_0 \in T \setminus N_c$. Then

$$c = \sigma_f(y_0) \in \{-1, 1\}.$$

This is the positive-measure sign-identification lemma with the same almost-everywhere equality and the hypothesis $\lambda(T) > 0$. Put $\varepsilon = c$.

2. **Integrate using finite target measure.** The relevant integrability calculation is

$$\int_T |1| d\lambda = \lambda(T) < \infty.$$

Thus the observable $\phi(y) = 1$ satisfies the hypothesis of signed transfer. With the same maps and sets, that formula and Step 1 give

$$\begin{aligned} \int_S J_f(x) d\lambda(x) &= \int_T \sigma_f(y) d\lambda(y) \\ &= \int_T c d\lambda(y) = c\lambda(T) = \varepsilon\lambda(T). \end{aligned}$$

The replacement of σ_f by c uses $\lambda(N_c) = 0$. The final integral is finite because T has finite measure.

Main citations

- Exact declaration and its source —
`MathlibAnnex.BilipschitzOrientation.integral_det_fderiv_eq_signed_volume`

- An almost-everywhere constant on the target —
MathlibAnnex.BilipschitzOrientation.targetJacobianSign_ae_const
- Signed transfer for the observable one —
MathlibAnnex.BilipschitzOrientation.signed_area_transfer
- Positive measure forces the constant to be a sign —
MathlibAnnex.BilipschitzOrientation.targetJacobianSign_const_is_pm_one
- Total Jacobian sign, including the zero convention —
MathlibAnnex.BilipschitzOrientation.domainJacobianSign
- Transport of the sign by the specified inverse —
MathlibAnnex.BilipschitzOrientation.targetJacobianSign

Lean source signature (exact)

```
theorem integral_det_fderiv_eq_signed_volume {n : ℕ}
  (D : BilipschitzOpenData n) (hpre : IsPreconnected D.target)
  (hpos : 0 < volume D.target) (hfinite : volume D.target ≠ ∞) :
  ∃ ε : ℝ, (ε = 1 ∨ ε = -1) ∧
    ∫ x in D.source, (fderiv ℝ D.f x).det ∂volume =
      ε * (volume D.target).toReal
```

In the source	Mathematical meaning
(D : BilipschitzOpenData n)	The record described in this Card: open $S, T \subseteq E = \mathbb{R}^n$, maps $f, g : E \rightarrow E$ restricting to inverse bijections $S \leftrightarrow T$, global upper Lipschitz bounds C_f, C_g , and global lower bound $\ell \ x - x'\ _\infty \leq \ f(x) - f(x')\ _\infty$ with $\ell > 0$. The source name D is not the derivative symbol.
D.source; D.target; D.f; D.g	The local mathematical objects S, T, f, g , respectively, throughout this Card.
(hpre : IsPreconnected D.target)	The separate target preconnectedness assumption.
(hpos : 0 < volume D.target)	The positive-measure assumption $0 < \lambda(T)$, used to identify an a.e. constant with one of the two sign values.
(hfinite : volume D.target ≠ ∞)	The separate finite-measure assumption $\lambda(T) < \infty$, used for integrability of the constant target weight.
∃ ε : ℝ, (ε = 1 ∨ ε = -1) ∧	The conclusion chooses one real sign $\varepsilon \in \{-1, 1\}$; it is not a prescribed input.
In the source	Mathematical meaning
∫ x in D.source, (fderiv ℝ D.f x).det ∂volume	The real integral $\int_S J_f(x) d\lambda(x)$ of the signed Jacobian of f .

In the source	Mathematical meaning
<code>ε * (volume D.target).toReal</code>	The equal real quantity $\varepsilon\lambda(T)$. Finiteness identifies the extended measure with its real value; this conversion is confined to the source expression.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 19–25:

```
noncomputable section

open Set MeasureTheory Filter
open scoped BigOperators ENNReal

namespace MathlibAnnex
namespace BilipschitzOrientation
```

▼ Exact content identity

Declaration: `MathlibAnnex.BilipschitzOrientation.integral_det_fderiv_eq_signed_volume`

Accepted content SHA-256: `f17cc47db523e628895b64c4138c01c916cd3b2cfaafb433e0e9f4993475de29`

Accepted source guide SHA-256: `d67ae7011cb95787bfef811e676e9a8d30b2f96d1175d1c5c57190cc8f907e54`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`